Evolution terms from genuine two ρ's

1 Group I (with c.c.)

1.1 Row I

Only 4 ρ's

1.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[U^{ab}(w)\mathbb{A}_i^{(3)\,b}(k^+,-w) - \bar{\mathbb{A}}_i^{(3)\,a}(k^+,-w)\right]$$

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{16\pi^4}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon} - 2\gamma - \log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x) - U^{ab}(x)\,\rho^b(x)\right] \tag{1.1}$$

1.3 Row III

Only 3 ρ's

1.4 Row IV

Only 3 ρ's

1.5 Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_{x,z}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[-\bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+,w-z;p^+,z-x)\bar{\mathbb{B}}^{bc}_{2jk}(k^+-p^+,-z;p^+,-x) + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+,w-z;p^+,z-x)\,U^{db}(x)\,U^{ce}(z)\right.$$
$$\left.\mathbb{B}^{be}_{2jk}(k^+-p^+,-z;p^+,-x)\right]$$

We write it in two parts:

$$\frac{d^3N_{2a\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.2}$$

$$\frac{d^3N_{2b\,\rm UV\,NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[U^{af}(y) - U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{1.3}$$

$$\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.$$
$$\left. - \frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right] \times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y) - N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$
$$- \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.$$
$$\left. + \frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right] \times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y) - N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$
$$- \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times$$
$$\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y) - N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y) \tag{1.4}$$

2 Group II

2.1 Row I

$$\frac{d^3N_{\text{NLO}}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big[\mathbb{B}^{\dagger\,bc'}_{2ji}(k^+,-w';p^+,-z)U^{c'a}(w') - \bar{\mathbb{B}}^{\dagger\,c'a}_{2ji}(k^+,-w';p^+,-z)U^{bc'}(z)\Big]\times$$
$$\Big[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Big]$$

$$\frac{d^3N_{\text{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2 - (x-w)^4\,(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left. + \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2 - \frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$
$$- f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Big] \tag{2.1}$$

2.2 Row II

Only 4 ρ's

2.3 Row III

Only 4 ρ's

2.4 Row IV

$$\frac{d^3N_{\text{NLO}}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y,y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y'-w';p^+,w'-z)$$
$$- \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y-w;p^+,w-z)$$
$$U^{bd}(z)U^{ce}(y)\mathbb{A}^{(1)\,e}_k(p^+ + k^+,-y) - U^{bd}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y-w;p^+,w-z)\bar{\mathbb{A}}^{(1)\,c}_k(p^+ + k^+,-y)\Big]$$

$$\frac{d^3N_{\text{NLO}}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{N_c}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon} - d_\perp\,\gamma_E\right.$$
$$\left. - \frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1\right]\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\times\Big[U^{ec}(y) - U^{ce}(x)\Big]\,\rho^{e'}(x')\rho^e(x) \tag{2.2}$$

3 Group III (with c.c.)

3.1 Row I

Only 3 ρ's

3.2 Row II

Only 3 ρ's

3.3 Row III

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+, -y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y' - w'; p^+, w' - z)$$
$$- \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+, -y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y' - w'; p^+, w' - z)U^{d'b}(z)\Big]\times\Big[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+, -w; p^+, -z)$$
$$- U^{\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+, -w; p^+, -z)\Big]$$

We write it in two parts:

3.3.1 First part:

Only 3 ρ's

3.3.2 Second part:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\Big[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times$$
$$\big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\big] \tag{3.1}$$

3.4 Row IV

Only 4 ρ's

3.5 Row V

Only 3 ρ's

3.6 Row VI

Only 3 ρ's